

HUMAIN



HUMAIN Create Studio

PREPARED BY  AAVYA

Product Requirements Document

Agentic, AI-Native Content and Experience Platform

Document type	Product Requirements Document (Customer Edition)
Prepared for	HUMAIN
Region	Kingdom of Saudi Arabia (KSA) — in-region hosting
Classification	Confidential — Client Presentation
Version	1.0
Date	June 2026

1. Executive Summary

HUMAIN Create Studio is a **bilingual (Arabic + English), agentic AI-native content and digital-experience platform** designed to let marketing, communications, editorial and brand teams produce on-brand, governed, multilingual content at a fraction of the cost and time of traditional content management systems.

At its core is a team of **specialised AI agents** that plan and carry out the multi-step work of content production — drafting, translating, optimising, brand-checking and assembling publish-ready experiences

— operating through a **governed, audited tool layer** so that every agent action is permission-scoped, logged and reversible, with humans approving anything that goes live.

The platform unifies two experiences over a **single content store and a single brand system**:

- **Create Studio** — a light, prompt-first surface where a person describes intent in natural language and the agents draft, translate, optimise and assemble publish-ready content.
- **Structured CMS** — a governed, enterprise-grade content backend providing the schema, roles, editorial workflow, versioning and audit trail demanded by a regulated, brand-sensitive organisation.

Both surfaces are powered by an **agentic AI backbone**: orchestrated agents that reason over the brief, retrieve relevant context (RAG), call governed content tools, and chain steps to a finished deliverable — every action governed by role-based permissions and a complete audit log.

The objective: reduce content production cost and cycle-time by an order of magnitude, while *raising* brand consistency, governance and Arabic-first quality — all hosted in-Kingdom under HUMAIN's control.

Snapshots of the working platform are in **Appendix A**.

2. Business Context and Drivers

Driver	Current-state challenge	Platform response
Arabic-first communication	Most CMS platforms treat Arabic/RTL as an afterthought; translation is manual and inconsistent	Native EN⇌AR with right-to-left rendering and AI-assisted, tone-preserving translation as a first-class capability
Content velocity	Producing a single bilingual page can take days across writers, translators, designers and reviewers	Prompt-to-draft in minutes; AI assists every step from brief to publish
Brand consistency	Brand drift across teams, agencies and channels	A central brand and design-token system that every piece of content inherits and is checked against
Governance and sovereignty	Sensitive content on foreign-hosted SaaS; weak audit trails	In-Kingdom, self-hosted deployment with role-based access, editorial gates and full audit logging
Cost of tooling	Per-seat SaaS licences + separate translation, SEO and DAM tools	One consolidated platform owned by HUMAIN, no per-seat lock-in

3. Vision and Goals

Vision: *Let anyone at HUMAIN turn an idea into a published, on-brand, bilingual digital experience — by describing it, not by operating a tool.*

Primary goals

- 1. **G1 — Speed:** Reduce average content production cycle-time by $\geq 70\%$.
- 2. **G2 — Bilingual quality:** Deliver publish-grade Arabic and English from a single workflow, with translation parity.
- 3. **G3 — Brand integrity:** Guarantee every published asset inherits approved brand tokens and guidelines.
- 4. **G4 — Governance:** Enforce who-can-do-what with a complete, queryable audit trail.
- 5. **G5 — Sovereignty:** Keep all content and AI processing under HUMAIN's control, in-Kingdom.

4. Business Impact and Value

4.1 Quantified impact (target outcomes)

Metric	Baseline (typical)	Target with Create Studio	Impact
Time to produce a bilingual page	2–4 days	2–4 hours	~85% faster
Translation turnaround (EN⇌AR)	1–2 days per asset	Minutes (AI draft + human review)	~90% faster
SEO metadata coverage	Manual / inconsistent	Auto-generated for 100% of content	Full coverage
Brand-compliance review effort	High, manual	Guided by system, exception-only	Major reduction
Tooling consolidation	3–5 separate tools	1 platform	Lower TCO + simpler ops
Content reuse	Low (copy-paste)	Block-based, versioned, reusable	Higher consistency

4.2 Strategic value

- **Sovereign capability** — a national-grade content platform owned and operated in-Kingdom, not rented from abroad.
- **Scalable team output** — small teams produce at large-team volume; specialists focus on judgement, not mechanics.

- **Compounding brand equity** — every asset is consistent by construction, strengthening brand over time.
 - **Defensible governance** — auditable, role-gated workflows suitable for regulated and high-visibility communications.
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5. Scope

5.1 In scope

- Prompt-first authoring (Create Studio) and structured CMS administration.
- Bilingual EN/AR content with RTL support.
- AI assistance: drafting, translation, summarisation, SEO, tagging, headline variants, FAQ generation, image alt-text, marketing copy.
- Block-based page composition and reusable content blocks.
- Editorial workflow, role-based access, versioning and audit logging.
- Brand and design-token management.
- Semantic + Arabic-aware search.
- Media management and AI-assisted/asynchronous video rendering.
- Secure delivery of published content to web channels.

5.2 Out of scope (this phase)

- Paid media / ad-buying automation.
 - Native mobile applications (content is delivered to responsive web).
 - Third-party social scheduling (available as a future integration).
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6. Personas and Roles

The platform is **role-governed**. Each persona sees only what their role permits. The model is **7 roles realising 9 personas** — the same role is specialised into distinct personas by **ABAC scope attributes** (site, department, and content locale) rather than by minting a separate role per scope. This keeps the permission surface small and auditable while still giving each persona a precise remit.

Persona	Role	Scope (ABAC)	Primary jobs-to-be-done
Platform Administrator	admin	Global, unrestricted	Configure platform, manage users and roles, oversee all content and settings
Site Administrator	siteAdmin	Site-scoped (sites)	Own a specific site/brand surface end-to-end — create, review and publish within that site; cannot manage users or global settings
Author (EN)	author	Locale = EN	Draft and edit English content with AI assistance
Author (AR)	author	Locale = AR	Draft and edit Arabic content with AI assistance
Reviewer	reviewer	Site-scoped	Move content through editorial gates; quality-check
Publisher	publisher	Site-scoped	Approve and publish to live channels
Brand Manager	brand	Global brand assets	Own brand guidelines and design tokens
Department Author (e.g. HR)	author	Department-scoped (department)	Produce department content within guardrails (e.g. HR owns Careers)
Viewer	viewer	Read-only	Read-only access to published content and dashboards

The 7 roles are `admin`, `siteAdmin`, `author`, `reviewer`, `publisher`, `brand`, and `viewer`. The 9 personas above are these roles combined with scope attributes — e.g. *Author (EN)* and *Author (AR)* are both the `author` role distinguished by their **locale** scope, and *Department Author* is `author` distinguished by its **department** scope. `siteAdmin` is the one genuinely distinct authority introduced for the Site Administrator persona: site-scoped admin powers (create / review / publish within its sites) without platform-level user, role, or global-settings control, so it cannot escalate privileges. **All scope attributes are access-enforced, not advisory:** site and department scopes gate read/write at collection and field level, and **locale scope is a hard write boundary** — an author assigned a locale (e.g. EN) cannot create or edit content in another locale (AR), and vice-versa (empty = all locales). This makes every one of the nine personas a distinct, enforced authorization profile.

7. Functional Requirements

7.1 Create Studio — Prompt-First Authoring

- **FR-1** The system **shall** let a user generate publish-ready drafts from a natural-language brief.
- **FR-2** The system **shall** provide guided AI actions on any content: *summarise, translate, generate SEO, suggest headlines, generate FAQs, write alt-text, draft marketing copy, extract tags*.
- **FR-3** The system **shall** preview content live in both English and Arabic, including correct RTL layout.

- **FR-4** The system **shall** present a unified workspace covering Projects, Content, Brand, Design System, and Search.

7.2 Structured Content Management

- **FR-5** The system **shall** model content as governed collections (pages, articles, blog posts, FAQs, sites, media, brand guidelines, users).
- **FR-6** The system **shall** compose pages from **reusable, versioned content blocks**: Hero, Rich Text, Media, CTA, Cards, FAQ, Stats, Quote, Gallery, Embed, Features, Steps, and Logos.
- **FR-7** The system **shall** autosave drafts and retain a version history per document, enabling rollback.
- **FR-8** Content **shall** be scoped to a *site*, so teams only touch the surfaces they own (attribute-based access).

7.3 Bilingual and Localisation

- **FR-9** Every content field **shall** support per-locale values (EN/AR) with independent draft/published state.
- **FR-10** The system **shall** provide AI-assisted EN⇌AR translation that preserves meaning and tone, always followed by human review.
- **FR-11** The delivery layer **shall** render Arabic right-to-left with correct typography.
- **FR-11a** The **platform UI itself shall be switchable between English and Arabic** — one click, with full right-to-left layout and Arabic typography — independently of the content locale.

7.4 Editorial Workflow and Governance

- **FR-12** Content **shall** progress through an editorial gate: **Draft** → **In Review** → **Approved**, with only authorised roles able to advance it.
- **FR-13** Publishing **shall** be permission-gated; unauthorised users cannot push content live and are guided to save as draft or request review.
- **FR-14** Every create/update/delete/publish action **shall** be recorded in an immutable **audit log**.
- **FR-14a** Access control **shall** combine **RBAC (7 roles)** with **ABAC** (site, department, locale) at both collection and field level; roles and access scope **shall** be editable by platform admins only, so a user cannot escalate their own privileges, and a `siteAdmin` is confined to its assigned sites and excluded from global settings.
- **FR-14b** The system **shall** provide a **user and access administration** screen showing each account's roles, department, ABAC scope, active status and **creation date**, with an at-a-glance RBAC capability matrix.

7.5 Brand and Design System

- **FR-15** The system **shall** maintain centrally managed **brand guidelines** and **design tokens** (colours, type, spacing) shared by all surfaces.
- **FR-16** The system **shall** offer AI-assisted brand-guideline suggestions to accelerate brand setup and evolution.

7.6 Intelligent Search

- **FR-17** The system **shall** provide search across all content, including **Arabic-aware** and **semantic (meaning-based)** retrieval.

7.7 Media and Video

- **FR-18** The system **shall** manage media assets with AI-generated descriptive alt-text for accessibility.
- **FR-19** The system **shall** support **asynchronous video rendering** with status tracking.

7.8 Agent and Integration Surface

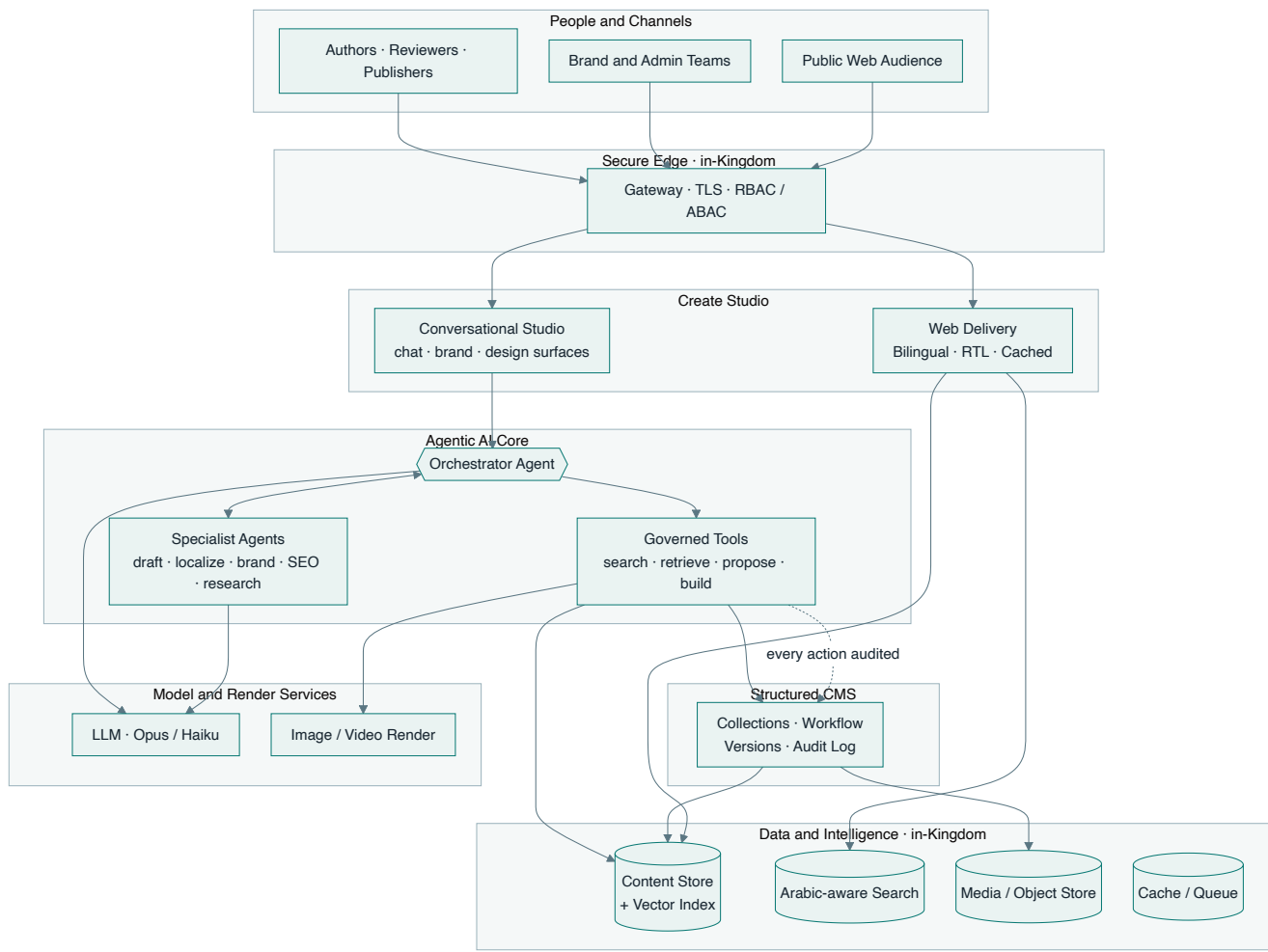
- **FR-20** The system **shall** expose a governed, audited tool interface (search / retrieve / propose content) for approved AI agents and downstream systems.

8. Non-Functional Requirements

Category	Requirement
Sovereignty	All content, data and AI processing hosted in-Kingdom (KSA) , under HUMAIN's control.
Security	Role-based + attribute-based access control; secure cookies; CORS/CSRF protection; secrets isolated from content.
Auditability	Complete, queryable audit trail of all content actions.
Localisation	First-class EN + AR (RTL); Arabic-aware search and translation.
Performance	Sub-second authoring interactions; cached, incrementally-rendered public delivery.
Scalability	Horizontally scalable services; queue-backed background processing for heavy jobs.
Reliability	Stateless application tier; managed data tier with health monitoring.
Extensibility	Provider-abstracted AI layer — model choice can evolve without re-platforming.
Observability	Centralised logs, metrics and tracing for operations.

9. Proposed Solution Architecture

9.1 High-level architecture



9.2 Logical components

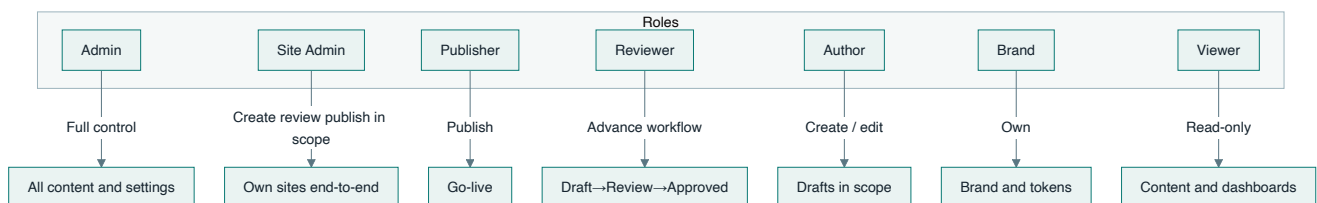
Layer	Responsibility
Create Studio	Natural-language authoring, AI actions, live bilingual preview
Structured CMS	Content schema, roles, editorial workflow, versioning, audit
AI Service	Drafting, translation, SEO, tagging, summarisation, semantic indexing
Web Delivery	Fast, cached, RTL-correct public rendering
Content Store	Single source of truth + vector embeddings for semantic search
Search Index	Arabic-aware and semantic retrieval
Cache / Queue	Performance + reliable background processing (e.g. video, indexing)
Object Store	Media and rendered assets, in-region

10. Content Lifecycle



Every transition is **role-gated** and **logged**.

11. Role and Permission Model



- **7 roles → 9 personas:** `admin`, `siteAdmin`, `author`, `reviewer`, `publisher`, `brand`, `viewer` cover all nine personas (see §6); locale and department personas are the `author` role differentiated by ABAC scope.
- **Site scoping (ABAC):** site admins, authors and editors act only on the sites assigned to them (empty = all).
- **Department scoping:** departmental authors are confined to their department's content.
- **No privilege escalation:** `siteAdmin` has site-scoped editorial authority but cannot manage users, roles or global settings — only `admin` can.

12. Agentic AI Platform

The platform's intelligence is delivered by a team of **specialised AI agents**, each owning a step of the content lifecycle and coordinated by an **orchestrator**. Agents are goal-directed and multi-step: given a brief or objective, they plan the work, retrieve relevant context, act through governed tools, check their own output, and hand off to one another until a publish-ready deliverable exists — with a human approving anything that goes live.

12.1 What makes it agentic

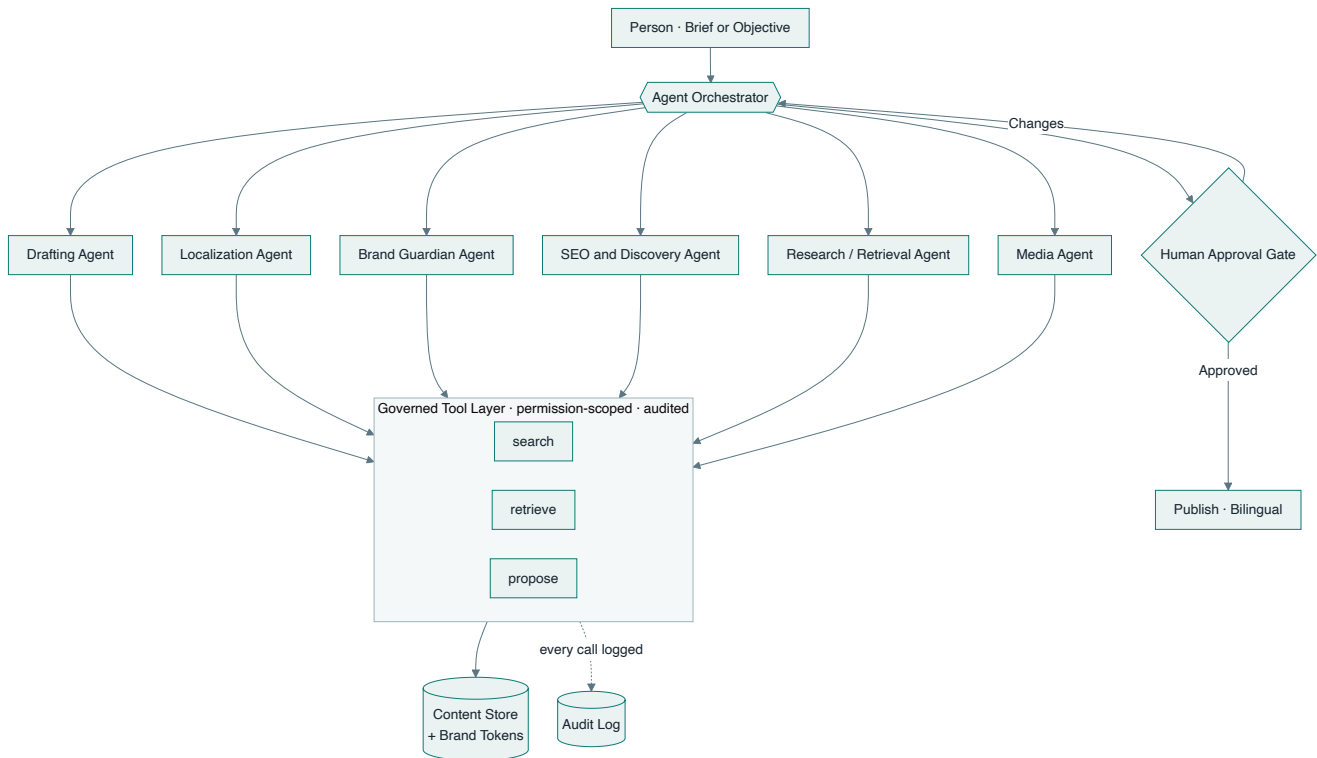
- **Goal-directed** — agents work from an objective ("produce a bilingual launch page from this brief"), not a single canned prompt.
- **Multi-step and self-checking** — they plan → retrieve → act → verify, iterating until the result meets brand and quality criteria.

- **Tool-using** — agents act only through a **governed tool layer** (search, retrieve, propose) rather than touching data directly.
- **Context-aware (RAG)** — they ground output in HUMAIN's own content and brand tokens via semantic, Arabic-aware retrieval.
- **Collaborative** — specialised agents hand off to each other under the orchestrator's plan.
- **Governed** — every agent runs under the same role/permission model as people, every action is audited, and **agents propose** — **humans dispose**.

12.2 Agent catalogue

Agent	What it does	Acts through
Orchestrator	Interprets the brief, plans the steps, delegates to specialist agents, assembles the final deliverable	Plan + governed tools
Drafting Agent	Turns a brief into a structured, block-based draft	<code>content_propose</code>
Localization Agent	Produces EN⇌AR translations with tone and meaning parity	<code>content_get</code> / <code>propose</code>
Brand Guardian Agent	Checks tone and visual tokens against brand guidelines; flags drift before review	Brand tokens + <code>content_get</code>
SEO and Discovery Agent	Generates metadata, keywords, taxonomy tags and internal-link suggestions	<code>content_search</code> / <code>propose</code>
Research / Retrieval Agent	Grounds output in HUMAIN's own content via semantic, Arabic-aware retrieval (RAG)	<code>content_search</code>
Media Agent	Generates descriptive alt-text and orchestrates image/video rendering	Media + render tools
Editorial Agent	Summarises changes, prepares review notes, routes content through the workflow gates	Workflow + <code>content_get</code>

12.3 Orchestration and governance



- Agents operate under the **same RBAC/ABAC** as human users — an agent can only touch content its scope permits.
- Agents **propose**; they **cannot publish**. A human approval gate stands before go-live.
- **Every agent tool call is written to the audit log**, giving a complete, queryable record of what each agent did and why.
- The agent runtime is **provider-abstracted** — HUMAIN is never locked to a single model vendor, and the strongest model can be assigned per task.

12.4 What the agent builds

From one conversation, the agent decides what is needed and **builds a complete, interactive, usable outcome** — not just a suggestion of copy:

Outcome	What the user gets
Website / landing page / app	A complete, responsive site built and previewed live in chat , with Open / Download
Structured content	Article, blog, page, press release, event, product, case study, FAQ, email, campaign or social post — an editable card that can be published to a CMS module (as a draft for review) or downloaded
Video	A production-ready script / storyboard with a one-click render to a real video
Image	A generated image , rendered inline
Brand guideline	A full guideline to verify, publish as the active brand the agents follow , or download
Design theme	A complete colour / type / radius theme with live preview, applied to the platform

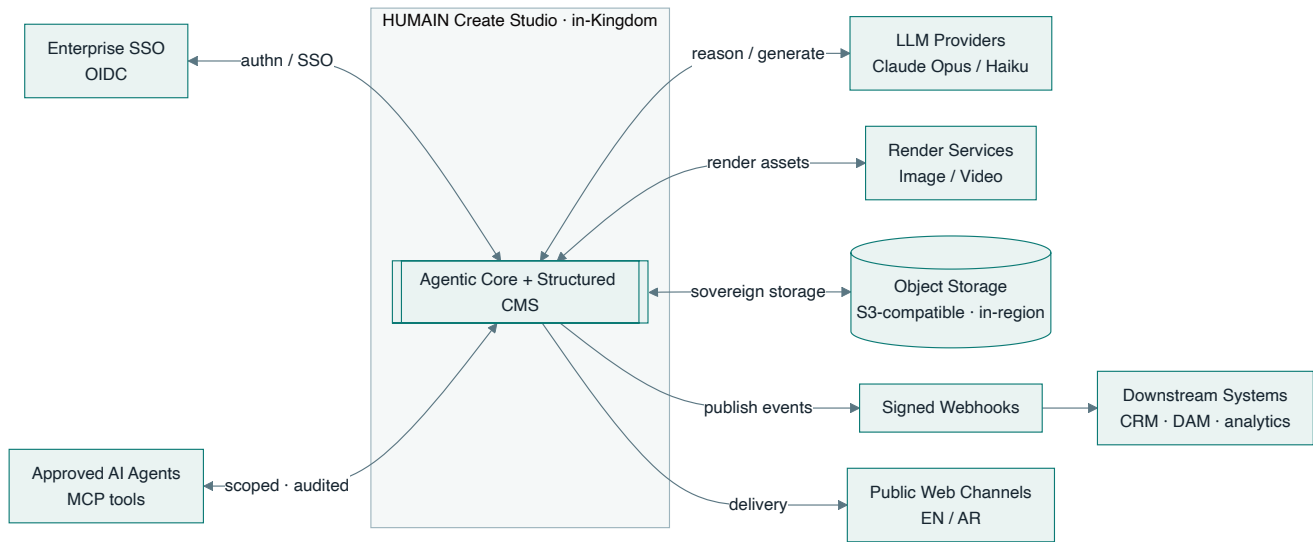
The agent draws on direct assist skills too — each also a one-click action: **drafting, EN⇌AR translation, SEO metadata, summarisation, headline variants, FAQ generation, taxonomy tagging, image alt-text and semantic, Arabic-aware search.**

Agents act; humans decide. No content reaches the public without human approval, and every agent action is permission-scoped and logged.

13. Integration Requirements

Integration	Purpose
Enterprise SSO (OIDC)	Single sign-on with HUMAIN identity
Object storage (S3-compatible, in-region)	Sovereign media and asset storage
Webhooks (signed)	Notify downstream systems on publish / content events
Agent tool interface	Governed programmatic access for approved AI agents
Web channels	Deliver published, bilingual content to public sites
LLM and render services	Provider-abstracted models (Opus/Haiku) + image/video rendering

13.1 Integration architecture

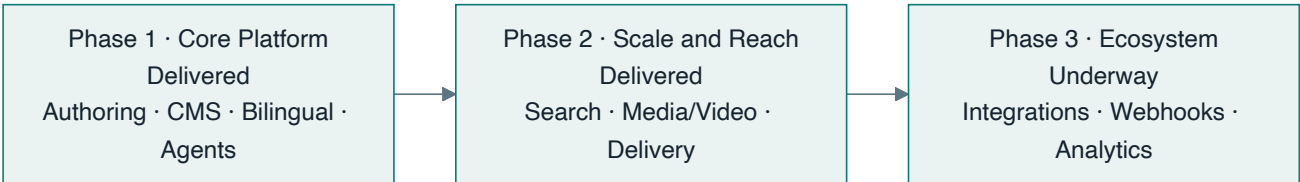


14. Security, Compliance and Sovereignty

- **In-Kingdom hosting** — data residency under HUMAIN's control.
- **Role-based + attribute-based access control** at every layer.
- **Editorial gating** prevents unreviewed or unauthorised publishing.
- **Immutable audit log** of all content actions for accountability.
- **Secrets isolation** — credentials and keys separated from content and code.
- **Secure transport** — encrypted (HTTPS/TLS) across all surfaces; secure session cookies; CORS/CSRF protections.

15. Delivery Roadmap and Status

The core programme (Phases 1–2) is **delivered and live** in-Kingdom; Phase 3 is **underway** (the agentic build set and MCP tool surface are already shipped).



Phase	Outcome	Focus	Status
Phase 1 — Core Platform	Teams author, translate and govern bilingual content	Studio, CMS, RBAC, workflow, agentic AI, brand tokens	Delivered
Phase 2 — Scale and Reach	Content is discoverable and richly delivered	Arabic-aware semantic search, media/video, cached public delivery	Delivered
Phase 3 — Ecosystem	Platform extends across HUMAIN's stack	Agent/MCP integrations, signed webhooks, advanced analytics	Underway

16. Success Metrics (KPIs)

KPI	Target
Bilingual content cycle-time reduction	≥ 70%
SEO metadata coverage	100% of published content
Brand-compliance exceptions at publish	Trending to near-zero
Translation review effort	≥ 80% reduction vs. manual
Audit completeness	100% of content actions logged
Author satisfaction (time-to-publish)	Measurable improvement quarter-on-quarter

17. Assumptions and Dependencies

- HUMAIN provides in-region infrastructure (or approves the proposed hosting) and an OIDC identity provider.
- Content owners define the initial brand guidelines and design tokens.
- AI model access is provisioned within HUMAIN's approved policy.
- Human review remains in the loop for all AI-assisted output prior to publish.

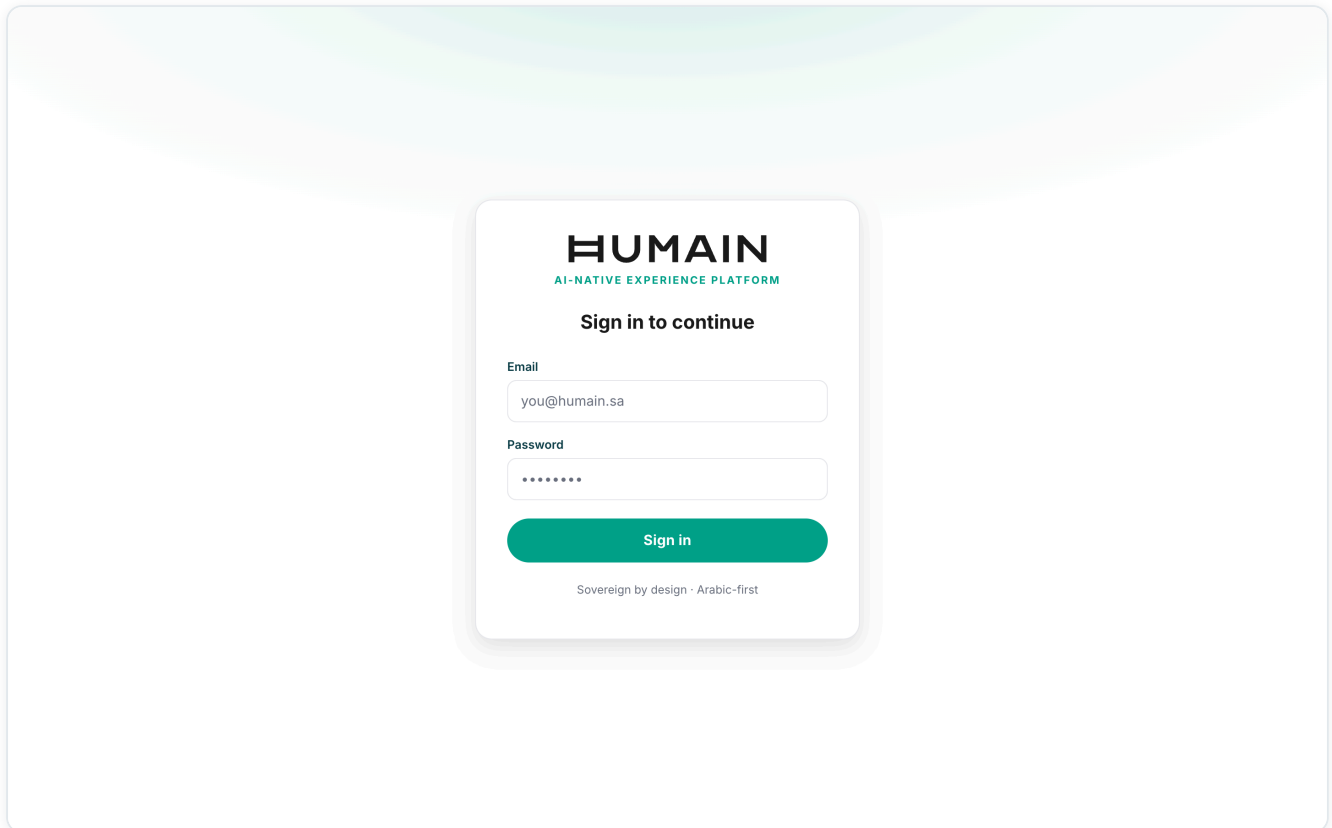
18. Glossary

Term	Meaning
Create Studio	The prompt-first authoring experience
Block	A reusable, versioned content component used to compose pages
Design tokens	Centralised brand values (colour, type, spacing) shared across surfaces
RTL	Right-to-left layout, required for Arabic
RBAC / ABAC	Role-based / attribute-based access control
Semantic search	Meaning-based retrieval beyond keyword matching
Audit log	Immutable record of all content actions
In-Kingdom	Hosted and processed within Saudi Arabia for data sovereignty

Appendix A — Product Snapshots

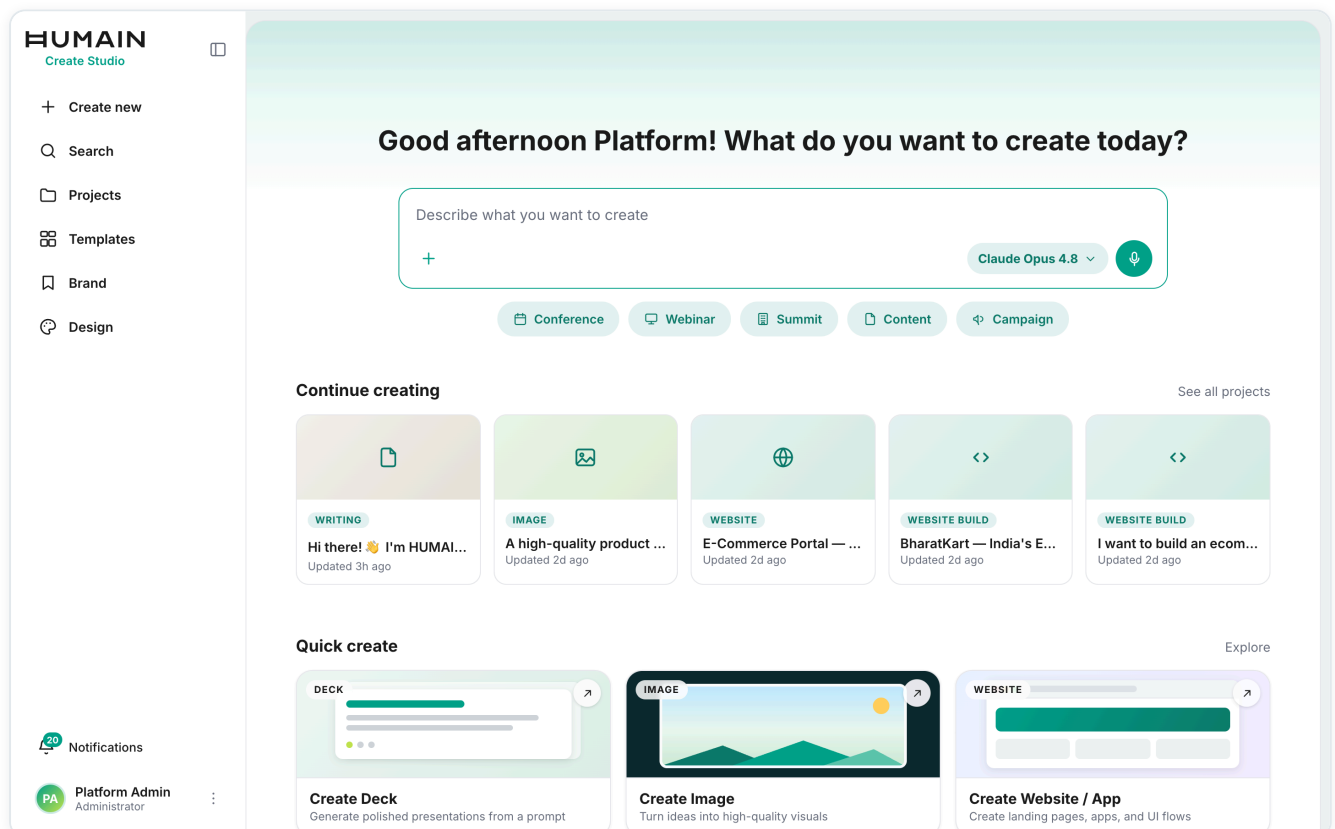
The following snapshots are taken from the working HUMAIN Create Studio platform, showing the realized capabilities described above.

A.1 Secure bilingual sign-in



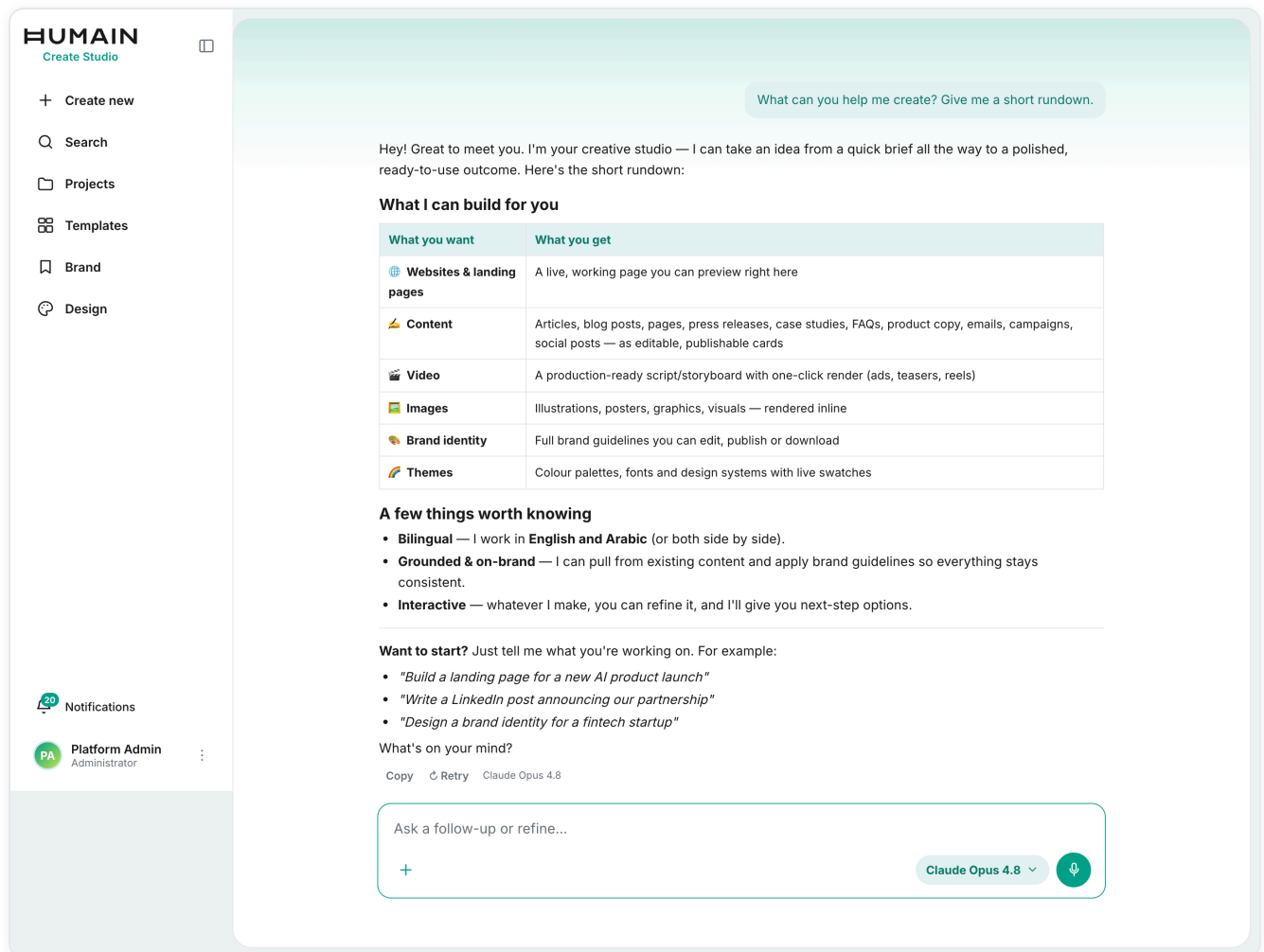
Branded, HTTPS sign-in backed by role-based access (OIDC-ready). Every session is permission-scoped.

A.2 Create Studio — one prompt box for everything



A single conversational entry point. The agent decides what to build; recent work and quick-create paths sit alongside.

A.3 Conversational, on-brand assistant



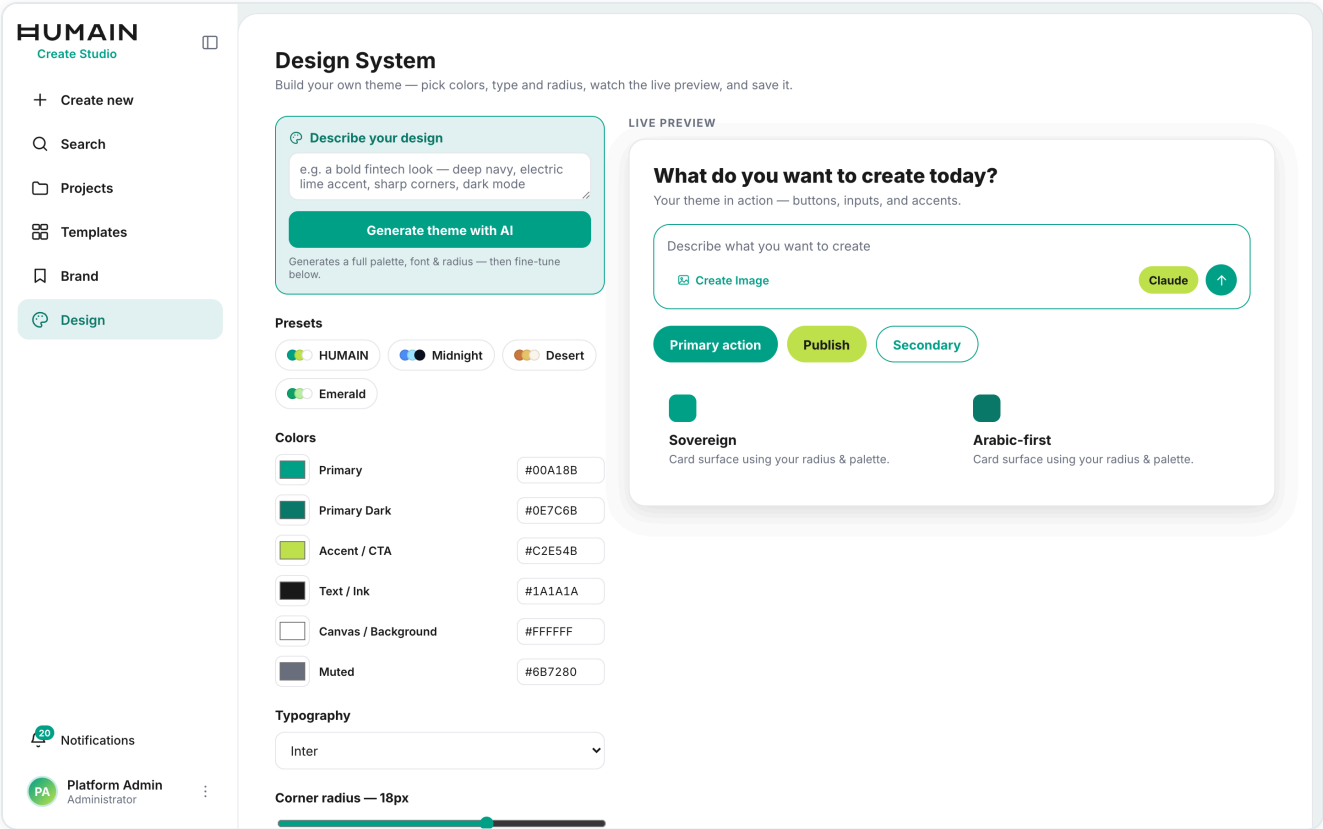
Natural, streaming replies (not form dumps) with rich formatting — the assistant explains what it can build, recommends next steps, and works bilingually (EN/AR), grounded in the active brand.

A.4 Build a website — live, in-chat preview

The screenshot displays the HUMAIN Create Studio interface. On the left is a sidebar with navigation options: 'Create new', 'Search', 'Projects', 'Templates', 'Brand', and 'Design'. Below these are 'Notifications' and a user profile for 'Platform Admin Administrator'. The main workspace shows a conversation with an AI agent. The agent's message includes a prompt to build a landing page for the HUMAIN AI Summit in Riyadh, followed by a 'WEBSITE' section describing the site's features and a list of tasks: 'Translate to Arabic', 'Add real speakers & sponsors', and 'Wire up registration'. Below the text is a 'LIVE PREVIEW' of the website. The preview shows a dark-themed landing page with the HUMAIN logo, the text 'RIYADH · 2025', and a large headline 'Where Enterprise Meets Intelligence'. It also includes a sub-headline, a paragraph of text, and two buttons: 'Reserve Your Seat →' and 'View Agenda'. At the bottom of the preview are 'Copy', 'Retry', and 'Claude Opus 4.8' options. Below the preview is a text input field for follow-up questions, a plus sign icon, and a 'Claude Opus 4.8' button with a speech icon.

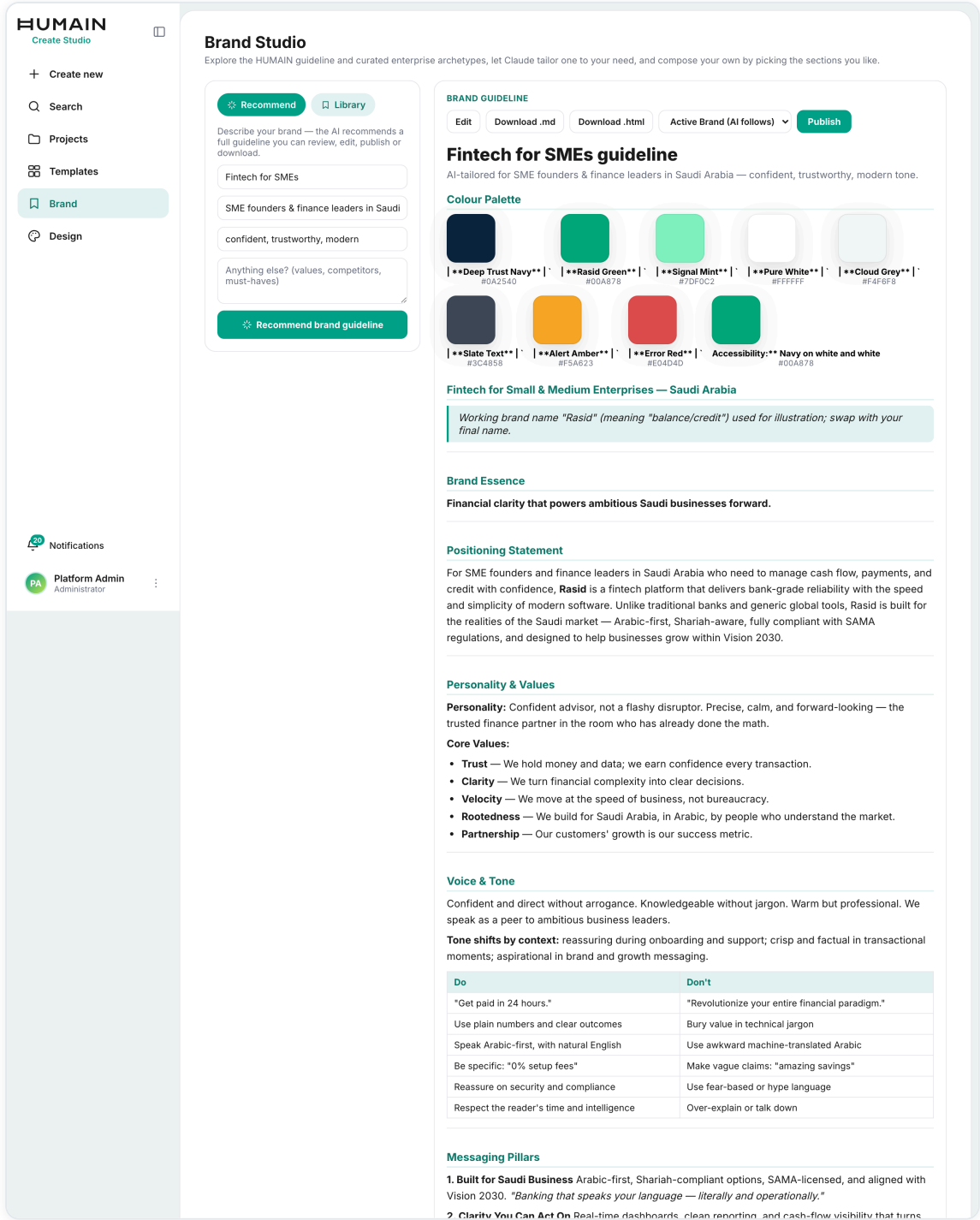
Ask for a site and the agent **builds it**, rendered live in the conversation with Open / Download — not a description. The same pattern produces editable, publishable content, images and video.

A.5 Design System — prompt-driven, agent-governed theme



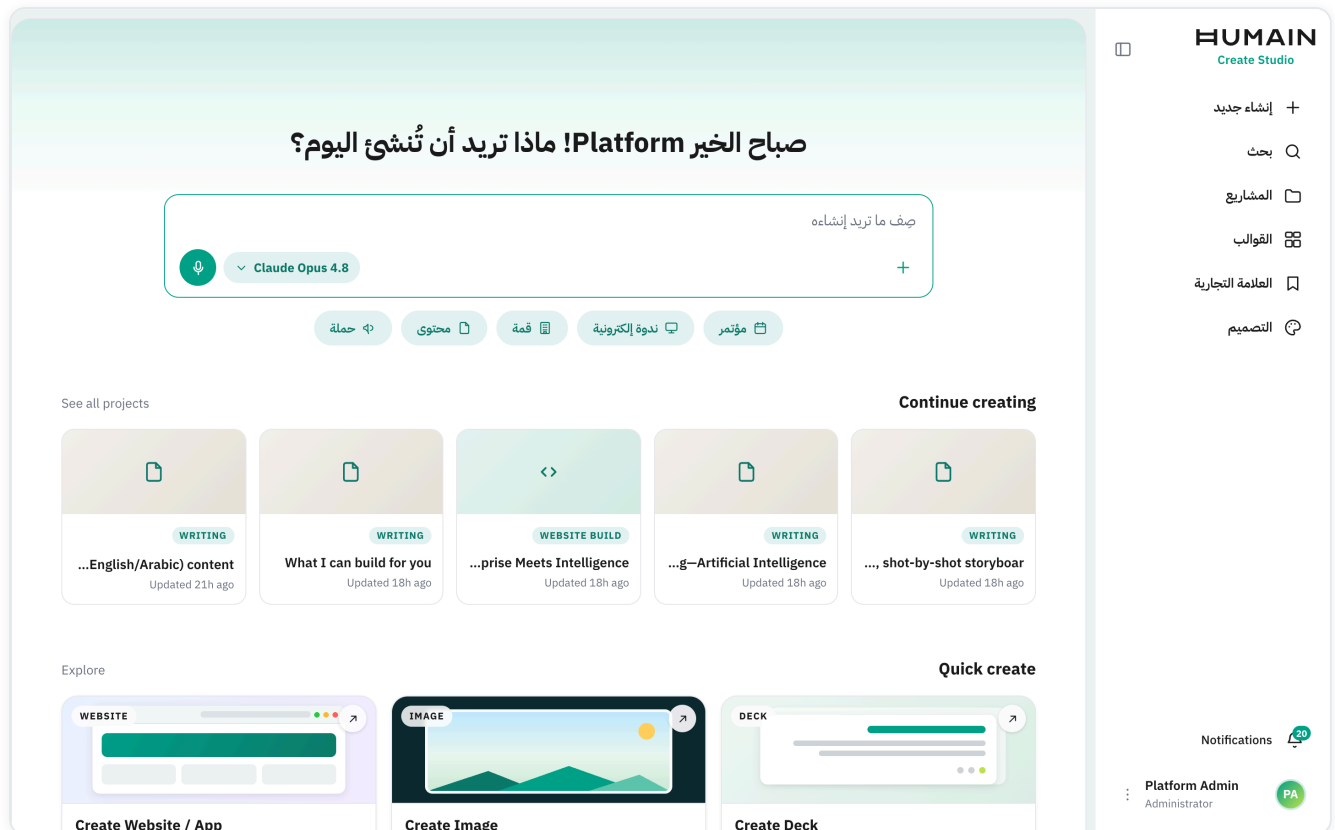
Describe a look and the agent generates a complete theme (palette, type, radius, light/dark) with a live preview; tokens are shared across every surface, EN + AR.

A.6 Brand Studio — recommend → review → publish



The AI recommends a full brand guideline to review and verify, then publish as the **active brand the agents follow**, or download.

A.7 Bilingual platform UI — English / Arabic



A one-click language switch turns the entire platform UI to **Arabic with full right-to-left layout** (and back to English) — navigation, prompts and actions all localised. The same content store serves both languages.

Prepared for HUMAIN — Confidential. This document describes the HUMAIN Create Studio platform; Appendix A shows snapshots of the working product.